

Technology #90

Optimize pacemaker programming (e.g., increase sensing threshold) to mitigate oversensing

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: EARLY_CONCEPT
Generated: 2026-08-27

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Optimize pacemaker programming (e.g., increase sensing threshold) to mitigate oversensing

WHY DOES IT MATTER?

Pacemaker battery depletion requiring surgical replacement

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 2 governing equations, 3 design inputs, 14 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in rf_wireless. 4-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

canonical invention specification (with epistemic classes); engineering specification (this document, all blocks classed); design input/output graph with explicit IDs; killer-experiment definition and its EIG basis; evidence custody references and hashes; unresolved-uncertainty register; domain equation set (symbolic; sources identified)

WHAT DOES THE BUYER HAVE TO BUILD?

actual geometry and detailed design; sourced engineering parameters and acceptance thresholds; physical prototypes; all verification and validation results; manufacturing process qualification; regulatory submissions

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Research Partnership

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 2 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 4-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: EARLY_CONCEPT

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Implant two pacemakers in a controlled setting and measure battery life with and without optimized programming. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-04 - article for VF-002 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: run killer experiment: falsification_test_from_candidate (EIG/cost 0.44573539382492045). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 2 equations. Key assumptions: in-body path loss exceeds free-space — margin is MODEL_DERIVED; noise figure and interference terms must be budgeted explicitly. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

2 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Mechanically mediated device-device interaction causing oversensing and pacing inhibition

POTENTIAL VALUE DRIVER: Addresses: Pacemaker battery depletion requiring surgical replacement

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: rf_wireless

Disciplines: RF engineering, antenna design, signal processing, EMC

Architecture: PROPOSED architecture (no physical system exists): transmit chain + power limits; channel / tissue path (path loss model); receive chain + detection; regulatory exposure constraint (SAR)

Subsystems:

?: transmit chain + power limits - ?

?: channel / tissue path (path loss model) - ?

?: receive chain + detection - ?

?: regulatory exposure constraint (SAR) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Optimize pacemaker programming (e.g., increase sensing threshold) to mitigate oversensing

Key Physics: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

```
{'equation_id': 'RF-001', 'expression': 'P_r = P_t * G_t * G_r * (lambda / (4 * pi * d))^2', 'source':  
{'text': 'Friis transmission equation (standard antenna-theory relation)', 'epistemic_class':  
'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'far field, line of  
sight; tissue path requires additional loss term', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain':  
'rf_wireless'}, 'assumptions': ['in-body path loss exceeds free-space — margin is MODEL_DERIVED'],  
'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'reason': 'domain template rf_wireless declares  
this relation as part of its governing model', 'mechanism_mentions': []}}
```

```
{'equation_id': 'RF-002', 'expression': 'SNR = P_signal / P_noise', 'source': {'text': 'standard  
detection-theory quantity (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT',  
'verify_before_release': True}, 'applicability': {'condition': 'defined for the chosen detection  
bandwidth', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['noise  
figure and interference terms must be budgeted explicitly'], 'selection_rationale': {'epistemic_class':  
'MODEL_DERIVED', 'reason': 'domain template rf_wireless declares this relation as part of its governing  
model', 'mechanism_mentions': []}}
```

Assumptions:

- in-body path loss exceeds free-space — margin is MODEL_DERIVED
- noise figure and interference terms must be budgeted explicitly

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'BATTERY_FAILURE', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'adversarial dimension: unsupported_mechanism', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: weak_transfer', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: obvious_combination', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: prior_art', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: contradiction', 'basis': 'COMPUTED (attack engine verdict)'}

- {'regime': 'adversarial dimension: boundary_failure', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: engineering_infeasibility', 'basis': 'COMPUTED (attack engine verdict)'}
- {'regime': 'adversarial dimension: regulatory_incompatibility', 'basis': 'COMPUTED (attack engine verdict)'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Must maintain cardiac pacing for 5-10 years without surgical replacement	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Mechanically mediated device-device interaction causing oversensing and phrenic nerve stimulation	MODEL_HYP
DI-003	Expected effect target	Prevent premature battery depletion by avoiding unnecessary pacing inhibition	MODEL_HYP

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Intervention architecture realizing the proposed mechanism	ABSENT	optimize pacemaker programming (e.g., increase sensing and hold-off) mitigate oversensing
DO-002	Effect-delivery subsystem achieving the expected effect	ABSENT	effect magnitude targets (no sourced values exist); actuation or passive
DO-003	Constraint-compliance envelope	ABSENT	quantified constraint limits; compliance test mapping

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
carrier frequency	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
transmit power (regulatory)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
antenna gain and pattern	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
path length and depth	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
receiver noise figure	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
required BER at data rate	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
SAR at tissue interface	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
DO response for Intervention	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
DO response for Effect-del	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification
DO response for Constraint	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(to be assigned candidate) target value and a verification

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
BATTERY_FAILURE	the device failure that motivates the establishment	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Passive supported_mechanism	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Weak transfer from	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Passive combination	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Point intervention (programming)	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Passive contradiction	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Boundary failures (e.g., pass	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Passive engineering_infeasibility	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
adversarial dimension	Passive regulatory_incompatibility	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
link margin collapse	domain generic candidate failure	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
SAR exceedance at trans	domain generic candidate failure	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
detuning from dielectric	domain generic candidate failure	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
interference in clinic	domain generic candidate failure	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)
antenna feed joint fatigue	domain generic candidate failure	NOT ESTABLISHED (requires design work)	ENGINEERING_PROPOSED	UNKNOWN — untested candidate failure mode (Art. 2)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
BATTERY_FAILURE	the device failure that motivates the invention	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
adversarial dimension: unsupported_mechanism	PASS	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
adversarial dimension: weak_transfer	The transfer from the single case to a higher predictable design	NOT ESTABLISHED (requires design work)	UNKNOWN — superficial tested candidate failure mode (Art. XXV)
adversarial dimension: obvious_combination	PASS	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
adversarial dimension: prior_intervention	The intervention (programming optimization) is already specified in the design	NOT ESTABLISHED (requires design work)	UNKNOWN — making tested candidate failure mode (Art. XXV)
adversarial dimension: complex_condition	PASS	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
adversarial dimension: boundary_conditions	Boundary conditions (e.g., patient intrinsic conditions)	NOT ESTABLISHED (requires design work)	UNKNOWN — are vaguely defined in the industry standard (Art. XXV)
adversarial dimension: engineering_infeasibility	PASS	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
adversarial dimension: regulatory_incompatibility	PASS	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
link margin collapse at maximum power	domain-specific candidate failure mode	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
SAR exceedance at transmission	domain-specific candidate failure mode	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
detuning from dielectric loading of nearby tissue	domain-specific candidate failure mode	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
interference in clinic RF environment	domain-generic candidate failure mode	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)
antenna feed joint fatigue failure	domain-generic candidate failure mode	NOT ESTABLISHED (requires design work)	UNKNOWN — untested candidate failure mode (Art. XXV)

9. Verification Strategy

[illegible]

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification passes)	NOT ESTABLISHED	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible antenna dielectric	implant telemetry practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible antenna dielectric	NOT ESTABLISHED	1	biocompatible antenna dielectric	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- high-density PCB with chip-scale RF front end: domain registry manufacturing pattern - candidate only; no process qualification exists
- implant-grade antenna integration and feed sealing: domain registry manufacturing pattern - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + canonical invention specification (with epistemic classes)
- + engineering specification (this document, all blocks classed)
- + design input/output graph with explicit IDs
- + killer-experiment definition and its EIG basis
- + evidence custody references and hashes
- + unresolved-uncertainty register
- + domain equation set (symbolic; sources identified)

YOU MUST DEVELOP:

- > actual geometry and detailed design
- > sourced engineering parameters and acceptance thresholds
- > physical prototypes
- > all verification and validation results
- > manufacturing process qualification
- > regulatory submissions

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'ALL sourced engineering parameter values', 'reason': 'no measurement or physical observation exists'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implanting the prop	physical verification (0.5G RMS)	ISO 13779-1	first physical article	1000h (1000h)
WP-02	constraint-compliance envelope per method	constraint-compliance envelope per method	constraint-compliance envelope per method	NOT ESTABLISHED	constraint-compliance envelope per method	1000h (1000h)
WP-03	article for VF-001	per method	Implant two pacemakers	ISO 13779-1	article for VF-001	1000h (1000h)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED	ISO 13779-1	article for VF-002	1000h (1000h)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 2 governing equations and a 4-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.